CUSTOMER CHURN ANALYSIS

A Data Analysis and Machine Learning Project

Project Overview

An end-to-end analysis of telecommunications customer churn using Python, SQL, Machine Learning, and Power BI.

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Field: Data Science & Data Analytics

Tools & Technologies:

- Python
- SQL
- Machine Learning
- Power BI
- Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

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Turning Customer Data into Actionable Business Insights

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1. Introduction

Customer churn is one of the major challenges faced by telecommunications companies. It occurs when customers stop using a company's services, which can negatively affect revenue, customer relationships, and long-term business growth.

Understanding why customers leave is important because it helps companies identify patterns in customer behavior and develop strategies to improve customer retention.

This project focuses on analyzing a telecommunications customer churn dataset to explore the factors associated with customer churn. It uses data analysis, SQL, machine learning, and Power BI to examine customer behavior, identify important churn patterns, build predictive models, and present the findings through an interactive dashboard.

The overall goal is to gain useful insights into customer churn and provide data-driven recommendations that may help telecommunications companies improve customer retention.

2. Project Objectives

The main objective of this project is to analyze customer churn and identify the factors that influence customers' decisions to leave a telecommunications company.

The specific objectives are:

1. **Data Cleaning and Preparation:** Prepare the dataset by handling missing values, correcting data types, and transforming categorical variables into a suitable format for analysis.
2. **Exploratory Data Analysis (EDA):** Explore customer data to identify patterns, trends, and relationships between customer characteristics and churn.
3. **SQL Analysis:** Use SQL queries to examine customer segments, calculate churn rates, and identify groups with higher churn rates.
4. **Machine Learning:** Build and evaluate classification models to predict customer churn and compare their performance using appropriate evaluation metrics.
5. **Power BI Visualization:** Develop an interactive dashboard to present key performance indicators and customer churn patterns.
6. **Business Insights and Recommendations:** Interpret the findings and suggest data-driven actions that may help improve customer retention.

Through these objectives, the project aims to demonstrate how data analysis and machine learning can support better business decisions.

3. Dataset Description

This project uses a telecommunications customer churn dataset containing **7,043 customer records**. The dataset provides information about customers' demographic characteristics, subscribed services, account details, contract types, tenure, and charges.

The target variable is **Churn**, which indicates whether a customer has left the company.

The target variable contains two categories:

- **No:** 5,174 customers who stayed with the company.
- **Yes:** 1,869 customers who left the company.

The dataset shows that approximately **26.54%** of customers churned, while **73.46%** remained with the company.

The dataset includes several types of information:

- **Customer Information:** Gender, senior citizen status, partner status, and dependents.
- **Service Information:** Internet service, phone service, online security, technical support, and other subscribed services.
- **Account Information:** Contract type, payment method, and paperless billing.
- **Financial Information:** Monthly charges and total charges.
- **Customer Tenure:** The length of time customers have stayed with the company.

The customerID column was excluded from the machine learning features because it serves only as a unique customer identifier and does not provide meaningful predictive information.

The Churn variable is used as the target for the machine learning classification models developed in this project.

4. Data Cleaning and Preparation

Before performing the analysis and building machine learning models, the dataset was cleaned and prepared to ensure data quality and consistency.

The following steps were performed:

4.1 Data Inspection

The dataset was examined to understand its structure, identify data types, and check for missing values and non-numeric columns.

4.2 Handling Missing Values

The TotalCharges column was initially stored as an object data type. It was converted into a numeric format. During this process, 11 missing values were identified. These values were replaced with 0 because they belonged to customers with a tenure of zero months.

After this step, no missing values remained in the prepared dataset.

4.3 Categorical Data Encoding

Categorical variables were converted into numerical representations using **One-Hot Encoding**. This allowed machine learning algorithms to process categorical customer information.

The feature matrix increased from 19 columns to 30 columns after encoding.

4.4 Train-Test Split

The dataset was divided into training and testing sets using an 80/20 split:

- **Training set:** 5,634 records.
- **Testing set:** 1,409 records.

Stratification was used to preserve the original distribution of the target variable in both sets.

The final prepared dataset contained no missing values and was ready for exploratory data analysis and machine learning.

4.5 Feature Scaling

Numerical features were standardized using StandardScaler to bring them to a comparable scale. The scaler was fitted on the training data using fit_transform() and then applied to the testing data using transform(). This approach prevented data leakage and ensured that the testing data was transformed using the same scaling parameters learned from the training data. Feature scaling was particularly important for the Logistic Regression model.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand customer behavior and identify the factors associated with customer churn. Different customer attributes and service features were analyzed to compare churn rates across customer groups.

5.1 Churn by Contract Type

Contract type showed a strong relationship with customer churn:

- Month-to-month: 42.71% churn rate.
- One year: 11.27% churn rate.
- Two year: 2.83% churn rate.

Customers with month-to-month contracts were considerably more likely to leave than customers with longer-term contracts.

5.2 Churn by Internet Service

The churn rate varied across internet service types:

- Fiber optic: 41.89%.
- DSL: 18.96%.
- No internet service: 7.40%.

Fiber optic customers had the highest churn rate among these groups.

5.3 Churn by Payment Method

Electronic check users had the highest churn rate:

- Electronic check: 45.29%.
- Mailed check: 19.11%.
- Bank transfer (automatic): 16.71%.
- Credit card (automatic): 15.24%.

This suggests that customers using electronic checks may require further investigation.

5.4 Churn by Technical Support

Customers without technical support had a churn rate of 41.64%, compared with 15.17% among customers who had technical support.

This indicates that access to technical support may be associated with better customer retention.

5.5 Customer Tenure and Monthly Charges

The analysis revealed differences in tenure and monthly charges between customers who stayed and those who left:

Metric	Stayed	Churned
Average tenure (months)	37.57	17.98
Median tenure (months)	38	10
Average monthly charges	61.27	74.44

Customers who churned generally had shorter relationships with the company and higher average monthly charges.

5.6 Other Customer Characteristics

Additional analysis showed:

- Senior citizens had a higher churn rate (41.68%) than non-senior citizens (23.61%).
- Customers without partners had a higher churn rate (32.96%) than customers with partners (19.66%).
- Customers without dependents had a higher churn rate (31.28%) than customers with dependents (15.45%).
- Gender showed only a small difference in churn rates: 26.92% for female customers and 26.16% for male customers.

5.7 Key EDA Findings

The analysis identified several customer groups with relatively high churn rates. The highest rates included:

- Month-to-month customers with fiber optic service: 54.61%.
- Month-to-month customers using electronic checks: 53.73%.
- Fiber optic customers without technical support: 49.37%.

Overall, contract type, internet service, payment method, technical support, customer tenure, and monthly charges showed notable relationships with customer churn.

6. SQL Analysis

SQL was used to analyze customer churn from different business perspectives. The analysis focused on grouping customers, calculating churn rates, and identifying customer segments with higher churn.

6.1 Customer Distribution by Contract Type

The first analysis examined the number of customers in each contract category:

Contract Type	Total Customers
Month-to-month	3,875
One year	1,473
Two year	1,695

Month-to-month contracts represented the largest customer group.

6.2 Churned Customers by Contract Type

SQL queries were used to count the customers who left the company in each contract category:

Contract Type	Churned Customers
Month-to-month	1,655
One year	166
Two year	48

The results showed that most churned customers belonged to the month-to-month contract group.

6.3 Churn Rate by Contract Type

The churn rate was calculated for each contract category using conditional aggregation:

Contract Type	Churn Rate
Month-to-month	42.71%
One year	11.27%
Two year	2.83%

These results confirmed that month-to-month customers had the highest churn rate.

6.4 Churn Rate by Payment Method

SQL was also used to compare churn rates across payment methods. Electronic check users had the highest churn rate at 45.29%, followed by mailed check users at 19.11%, automatic bank transfer users at 16.71%, and automatic credit card users at 15.24%.

6.5 Churn Analysis by Contract and Payment Method

A combined analysis was performed by grouping customers according to both contract type and payment method.

The highest churn rate was found among customers with month-to-month contracts who used electronic checks:

- Total customers: 1,850.
- Churned customers: 994.

- Churn rate: 53.73%.

This analysis helped identify a customer segment that may deserve additional attention in customer retention efforts.

6.6 SQL Analysis Summary

SQL provided a structured way to examine customer groups and calculate important business metrics. The results supported the findings from the exploratory data analysis and helped identify customer segments with higher churn rates.

7. Machine Learning

Machine learning classification models were developed to predict whether a customer would leave the telecommunications company or remain with it.

The target variable was Churn, which contains two classes: Yes and No.

7.1 Model Development

Two classification algorithms were trained and evaluated:

- **Logistic Regression**
- **Random Forest**

Both models were trained using the prepared dataset and evaluated on the testing set.

7.2 Model Evaluation

The models were evaluated using several classification metrics:

- **Accuracy:** Measures the overall percentage of correct predictions.
- **Precision:** Measures how many customers predicted to churn actually churned.
- **Recall:** Measures how many actual churned customers were correctly identified.
- **F1-score:** Combines precision and recall into a single metric.
- **ROC-AUC:** Measures the model's ability to distinguish between the two classes.

The evaluation results were:

Metric	Logistic Regression	Random Forest
Accuracy	80.62%	78.80%
Precision	65.73%	62.50%
Recall	56.42%	50.00%
F1-score	60.72%	55.60%
ROC-AUC	0.842	0.825

Logistic Regression achieved better results across all the reported evaluation metrics and was selected as the baseline model.

7.3 Confusion Matrix

The Logistic Regression model produced the following confusion matrix at the default classification threshold of 0.5:

	Predicted No	**Predicted Yes**
Actual No	925	110
Actual Yes	163	211

The model correctly identified 211 customers who churned, while 163 actual churned customers were incorrectly classified as staying.

7.4 Classification Threshold Adjustment

Since identifying customers who might leave was an important business objective, the classification threshold was adjusted from 0.5 to 0.3.

The results were:

Metric	**Threshold 0.5**	**Threshold 0.3**
Precision	65.73%	51.93%
Recall	56.42%	75.40%
F1-score	60.72%	61.50%

Reducing the threshold increased recall from 56.42% to 75.40%, meaning that the model identified a larger proportion of actual churned customers.

However, precision decreased because more customers were classified as likely to churn, including some who would actually stay.

7.5 Model Selection and Business Interpretation

Logistic Regression was selected because it achieved stronger evaluation results than Random Forest in this experiment.

Why Recall Is Important for This Project

Recall was considered an important evaluation metric because the main business objective is to identify as many customers at risk of churning as possible. A false negative occurs when the model predicts that a customer will stay, even though the customer actually leaves. Missing these customers may result in lost opportunities to take preventive retention actions. Therefore, improving recall helps the company identify more potential churn cases, even if this leads to some additional false positive predictions. Accuracy alone may not fully reflect how effectively the model identifies customers who are likely to leave.

The threshold of 0.3 was considered useful for the business objective of identifying more customers at risk of leaving. Although it produced more false positive predictions, it improved the model's ability to detect actual churned customers.

This demonstrates the importance of selecting a classification threshold based on business priorities rather than relying only on accuracy.

8. Power BI Dashboard

Power BI was used to create an interactive dashboard to present the main findings of the customer churn analysis. The dashboard provides a clear overview of customer retention and churn patterns through key performance indicators, charts, and interactive filters.

8.1 Key Performance Indicators (KPIs)

Three main KPIs were displayed at the top of the dashboard:

KPI	Value
Total Customers	7,043
Churned Customers	1,869
Churn Rate	26.54%

These indicators provide a quick overview of the company's customer base and overall churn performance.

8.2 Dashboard Visualizations

The dashboard includes five main visualizations:

Churned vs. Retained Customers by Contract Type: Compares the number of customers who left and those who stayed across different contract types.

Churn Rate by Contract Type: Shows the percentage of customers who churned under each contract category.

Churn Rate by Internet Service Type: Compares churn rates among DSL, Fiber optic, and customers without internet service.

Churn Rate by Payment Method: Highlights differences in churn rates across the available payment methods.

Churn Rate by Tech Support: Compares churn rates between customers with and without technical support.

8.3 Interactive Filters (Slicers)

Three interactive slicers were added to allow users to explore the dashboard by selecting specific customer groups:

Contract

Internet Service

Payment Method

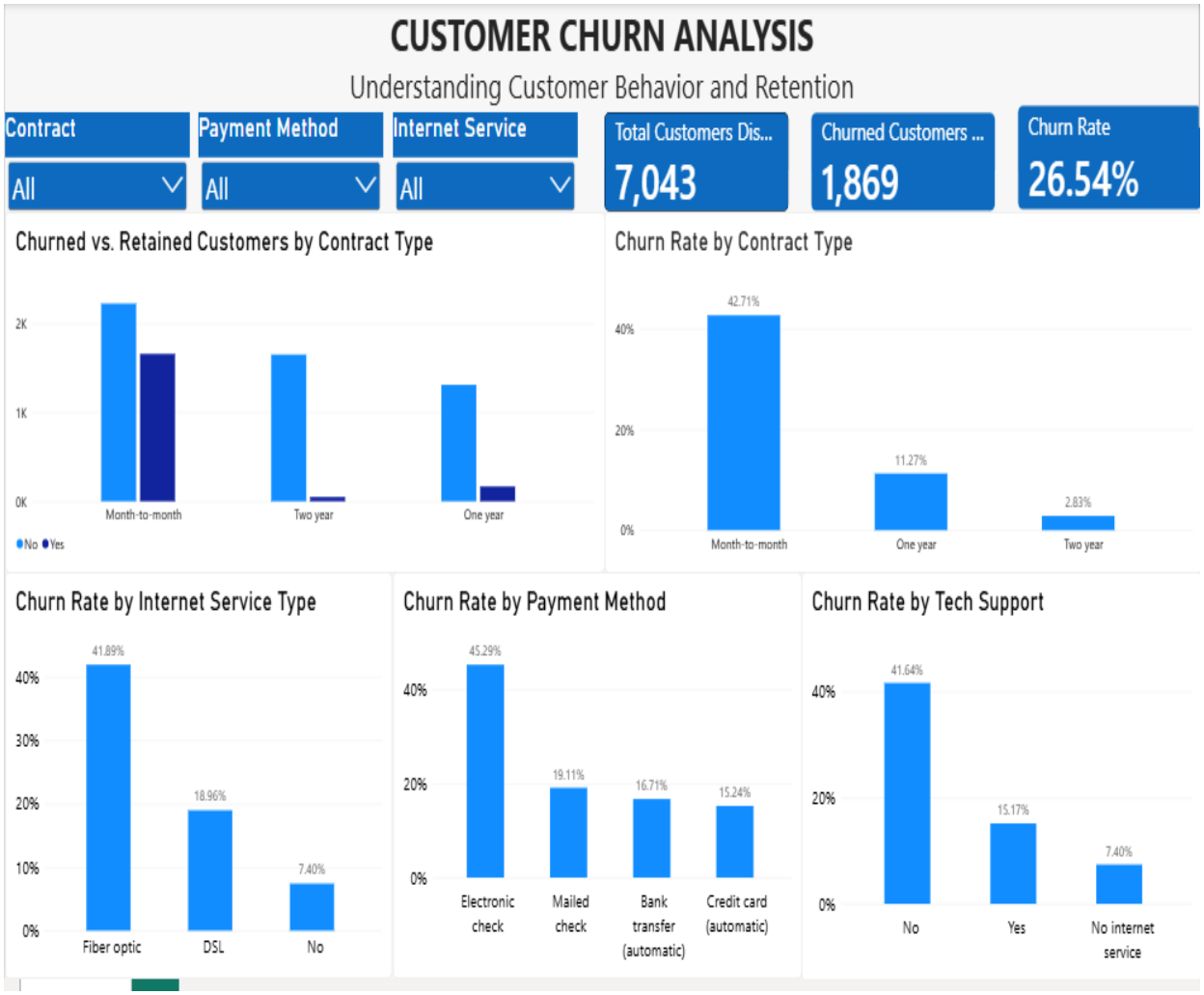
These filters update the dashboard visuals according to the selected categories, making it easier to investigate customer segments and compare their churn patterns.

8.4 Dashboard Summary

The Power BI dashboard brings together the key findings from the analysis in one interactive report. It allows users to monitor important churn indicators, explore customer segments, and identify groups with higher churn rates.

The dashboard supports data-driven decision-making by presenting complex analysis results in a clear and accessible visual format.

Figure 1. Interactive Customer Churn Analysis Dashboard.



9. Key Findings and Business Recommendations

9.1 Key Findings

Based on the exploratory data analysis, SQL queries, machine learning results, and Power BI dashboard, the project identified the following key findings:

1. **Contract Type:** Month-to-month customers had the highest churn rate at 42.71%, compared with 11.27% for one-year contracts and 2.83% for two-year contracts.
2. **Internet Service:** Fiber optic customers had a churn rate of 41.89%, making them the highest-churn group among the internet service categories.
3. **Payment Method:** Customers using electronic checks had the highest churn rate at 45.29%.
4. **Technical Support:** Customers without technical support had a higher churn rate (41.64%) than those with technical support (15.17%).

5. **Customer Tenure:** Customers who churned had a lower average tenure of 17.98 months, compared with 37.57 months for customers who stayed.
6. **Monthly Charges:** Customers who churned had higher average monthly charges (74.44) than customers who stayed (61.27).
7. **Machine Learning:** Logistic Regression achieved better overall evaluation results than Random Forest in this experiment. Adjusting its classification threshold to 0.3 increased recall to 75.40%.

9.2 Business Recommendations

Based on these findings, the following recommendations may help the company improve customer retention:

1. **Encourage Long-Term Contracts:** Develop suitable offers and incentives to encourage month-to-month customers to consider longer-term contracts.
2. **Investigate Fiber Optic Customer Churn:** Examine customer satisfaction, service quality, and pricing among fiber optic customers to better understand their high churn rate.
3. **Review Electronic Check Payments:** Investigate whether customers using electronic checks experience payment-related difficulties or other issues.
4. **Improve Technical Support:** Make technical support more accessible and encourage customers to use available support services.
5. **Focus on New Customers:** Develop early engagement and retention strategies for customers during their first months with the company.
6. **Review Pricing:** Evaluate monthly charges and consider suitable pricing options for customers who may be sensitive to service costs.
7. **Use Predictive Models:** Apply the Logistic Regression model to help identify customers who may be at risk of leaving, while reviewing its predictions before taking retention actions.

9.3 Overall Business Impact

The findings demonstrate how data analysis and machine learning can help identify customer groups associated with higher churn rates. By combining descriptive analysis, predictive modeling, and interactive visualization, the company can gain a better understanding of customer behavior and make more informed customer retention decisions.

10. Conclusion

This project explored customer churn in a telecommunications company using data analysis, SQL, machine learning, and Power BI.

The analysis identified several customer characteristics and service-related factors associated with higher churn rates, particularly month-to-month contracts, fiber optic internet service, electronic check payments, and the absence of technical support.

Machine learning models were developed to predict customer churn. Logistic Regression achieved better overall performance than Random Forest in this experiment. Adjusting the

classification threshold improved recall, which supported the project's objective of identifying more customers who might leave.

Finally, an interactive Power BI dashboard was created to present the main findings and allow users to explore customer segments.

Overall, this project provided practical experience in the data analysis workflow, from data cleaning and exploratory analysis to SQL querying, predictive modeling, and dashboard development. It also demonstrated how data-driven insights can support customer retention strategies and business decision-making.

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